

Polyak-Łojasiewicz Randomized Coordinate Descent (PL-COORD)

1 Algorithm Description

The algorithm iteratively updates the full parameter vector by computing the gradient along a randomly selected coordinate direction at each step. It uses this coordinate-wise gradient information to determine the update direction and step size for the entire parameter vector. It is designed for functions satisfying the Polyak-Łojasiewicz condition that are smooth with respect to individual coordinates.

2 Pseudo Code and Dry Run

2.1 Mathematical Formulation

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable function satisfying the Polyak-Łojasiewicz (PL) condition:

$$\frac{1}{2} \|\nabla f(\mathbf{w})\|^2 \geq \mu (f(\mathbf{w}) - f^*) \quad \forall \mathbf{w} \in \mathbb{R}^n,$$

where $\mu > 0$ is the PL parameter and f^* is the optimal value. Each coordinate-wise gradient $\frac{\partial f}{\partial w_i}$ is L_i -smooth:

$$\left| \frac{\partial f}{\partial w_i}(\mathbf{w} + s\mathbf{e}_i) - \frac{\partial f}{\partial w_i}(\mathbf{w}) \right| \leq L_i |s| \quad \forall s \in \mathbb{R}, i \in \{1, \dots, n\},$$

where $\mathbf{e}_i$ is the standard basis vector.

Update Rule: At iteration k , select a coordinate $i_k \in \{1, \dots, n\}$ uniformly at random. Update:

$$\mathbf{w}_k = \mathbf{w}_{k-1} - \eta \cdot \frac{\partial f}{\partial w_{i_k}}(\mathbf{w}_{k-1}) \cdot \mathbf{e}_{i_k},$$

where $\eta > 0$ is the step size.

Hyperparameters: - **Step Size** η (constant): Controls the magnitude of updates. Convergence requires $\eta < \frac{2}{L_{\max}}$, where $L_{\max} = \max_i L_i$. - **Sampling Strategy:** Uniformly random coordinate selection. - **Termination Conditions:** Maximum iterations $T_{\max}$ or gradient norm $\|\nabla f(\mathbf{w}_k)\| \leq \epsilon$.

2.2 Pseudocode

Initialize parameter vector $\mathbf{w}_0$, step size η , maximum iterations $T_{\max}$, tolerance ϵ . $k = 1$ to $T_{\max}$: Randomly select coordinate $i_k \in \{1, \dots, n\}$ uniformly. Compute partial derivative $g_k = \frac{\partial f}{\partial w_{i_k}}(\mathbf{w}_{k-1})$. Update parameter: $\mathbf{w}_k = \mathbf{w}_{k-1} - \eta g_k \mathbf{e}_{i_k}$. Check termination: $\|\nabla f(\mathbf{w}_k)\| \leq \epsilon$: Break loop. Return $\mathbf{w}_k$.

2.3 Dry Run Example

Objective Function: $f(w) = (w - 3)^2$ (scalar case $n = 1$). **Hyperparameters:** $w_0 = 0$, $\eta = 0.1$, $T_{\max} = 3$.

Iteration 1: 1. Gradient: $\frac{\partial f}{\partial w}(w_0) = 2(w_0 - 3) = 2(0 - 3) = -6$. 2. Update: $w_1 = w_0 - 0.1 \cdot (-6) = 0 + 0.6 = 0.6$. 3. Objective: $f(w_1) = (0.6 - 3)^2 = 5.76$.

Iteration 2: 1. Gradient: $\frac{\partial f}{\partial w}(w_1) = 2(0.6 - 3) = -4.8$. 2. Update: $w_2 = 0.6 - 0.1 \cdot (-4.8) = 0.6 + 0.48 = 1.08$. 3. Objective: $f(w_2) = (1.08 - 3)^2 = 3.6864$.

Iteration 3: 1. Gradient: $\frac{\partial f}{\partial w}(w_2) = 2(1.08 - 3) = -3.84$. 2. Update: $w_3 = 1.08 - 0.1 \cdot (-3.84) = 1.08 + 0.384 = 1.464$. 3. Objective: $f(w_3) = (1.464 - 3)^2 = 2.3593$.

Observation: The algorithm converges toward the minimizer $w^* = 3$, with the objective value decreasing monotonically.

3 Convergence Theory

3.1 Assumptions

We formalize the assumptions required for our convergence analysis as follows:

Assumption 1 (Coordinate-Wise Smoothness). *The objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable, and each coordinate directional derivative is L_i -smooth. That is, for all $\mathbf{x} \in \mathbb{R}^n$, $i \in \{1, 2, \dots, n\}$, and $t \in \mathbb{R}$,*

$$f(\mathbf{x} + t\mathbf{e}_i) \leq f(\mathbf{x}) + t\nabla_i f(\mathbf{x}) + \frac{L_i}{2}t^2,$$

where $\mathbf{e}_i$ is the standard basis vector and $\nabla_i f(\mathbf{x}) = \frac{\partial f}{\partial x_i}(\mathbf{x})$.

Assumption 2 (Polyak-Łojasiewicz (PL) Condition). *The function f satisfies the Polyak-Łojasiewicz inequality with constant $\mu > 0$: for all $\mathbf{x} \in \mathbb{R}^n$,*

$$\frac{1}{2}\|\nabla f(\mathbf{x})\|^2 \geq \mu(f(\mathbf{x}) - f^*),$$

where $f^* = \min_{\mathbf{x}} f(\mathbf{x})$ is the global minimum value.

Assumption 3 (Existence of Minimizer). *The function f attains its minimum at some $\mathbf{x}^* \in \mathbb{R}^n$, and $f^* = f(\mathbf{x}^*) > -\infty$.*

3.2 Main Convergence Theorem

We analyze the convergence of the randomized coordinate descent algorithm that, at each step, selects a coordinate $i_k \in \{1, \dots, n\}$ uniformly at random and updates the iterate via:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \frac{1}{L_{i_k}} \nabla_{i_k} f(\mathbf{x}_k) \mathbf{e}_{i_k}.$$

The following theorem establishes linear convergence in expectation.

Theorem 1 (Linear Convergence under PL and Coordinate Smoothness). *Suppose f satisfies Assumptions 1–3. Let $L_{\max} = \max_i L_i$. Then the sequence $\{\mathbf{x}_k\}$ generated by randomized coordinate descent satisfies:*

$$\mathbb{E}[f(\mathbf{x}_k) - f^*] \leq \left(1 - \frac{\mu}{nL_{\max}}\right)^k (f(\mathbf{x}_0) - f^*).$$

Proof. By coordinate-wise smoothness (Assumption 1), the one-step progress satisfies:

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - \frac{1}{2L_{i_k}} (\nabla_{i_k} f(\mathbf{x}_k))^2.$$

Taking expectation over $i_k \sim \text{Uniform}(\{1, \dots, n\})$:

$$\mathbb{E}[f(\mathbf{x}_{k+1}) \mid \mathbf{x}_k] \leq f(\mathbf{x}_k) - \frac{1}{2n} \sum_{i=1}^n \frac{1}{L_i} (\nabla_i f(\mathbf{x}_k))^2.$$

Since $\frac{1}{L_i} \geq \frac{1}{L_{\max}}$, we have:

$$\sum_{i=1}^n \frac{1}{L_i} (\nabla_i f(\mathbf{x}_k))^2 \geq \frac{1}{L_{\max}} \|\nabla f(\mathbf{x}_k)\|^2.$$

Applying the PL condition (Assumption 2) and rearranging:

$$\mathbb{E}[f(\mathbf{x}_{k+1}) - f^* \mid \mathbf{x}_k] \leq \left(1 - \frac{\mu}{nL_{\max}}\right) (f(\mathbf{x}_k) - f^*).$$

Iterating this inequality completes the proof. $\square$

3.3 Theoretical Properties

3.3.1 Stability and Hyperparameter Sensitivity

The convergence rate depends critically on the stepsize choice $1/L_i$. Overestimating L_i (i.e., using smaller stepsizes) preserves stability but slows convergence, while underestimating L_i risks divergence. The rate deteriorates with larger n and $L_{\max}$, akin to the condition number in gradient descent.

3.3.2 Comparison to Full Gradient Descent

Full gradient descent under the PL condition achieves a rate of $\mathcal{O}((1 - \mu/L)^k)$, where L is the global smoothness constant. Coordinate descent incurs an additional $\mathcal{O}(n)$ factor but reduces per-iteration cost from $\mathcal{O}(n)$ to $\mathcal{O}(1)$, making it preferable for high-dimensional problems.

3.3.3 Special Cases

- **Isotropic Smoothness:** If $L_i = L$ for all i , the rate simplifies to $(1 - \frac{\mu}{nL})^k$, reflecting uniform curvature.
- **Convex Functions:** If f is convex and μ -PL, the result still holds, though convexity is not required.

3.4 Discussion

The derived convergence theory shows that randomized coordinate descent achieves linear convergence for functions satisfying the PL condition and coordinate-wise smoothness. This generalizes classical results for gradient descent to settings where full gradients are unavailable or expensive. The rate depends on the interplay between the PL constant μ , the worst-case coordinate smoothness $L_{\max}$, and the dimension n . Practical implementations may benefit from importance sampling over coordinates (e.g., weighting by $1/L_i$) to improve the effective μ/L_{avg} ratio. Future work could extend these results to non-uniform sampling, inexact gradients, or adaptive stepsize rules.

4 Convergence Proofs

4.1 Preliminaries (Definitions and Lemmas)

4.1.1 Definitions

Assumption 1 (Coordinate-Wise Smoothness): The objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable, and each coordinate directional derivative is L_i -smooth. That is, for all $\mathbf{x} \in \mathbb{R}^n$, $i \in \{1, 2, \dots, n\}$, and $t \in \mathbb{R}$,

$$f(\mathbf{x} + t\mathbf{e}_i) \leq f(\mathbf{x}) + t\nabla_i f(\mathbf{x}) + \frac{L_i}{2}t^2,$$

where $\mathbf{e}_i$ is the standard basis vector and $\nabla_i f(\mathbf{x}) = \frac{\partial f}{\partial x_i}(\mathbf{x})$.

Assumption 2 (Polyak-Łojasiewicz (PL) Condition): The function f satisfies the Polyak-Łojasiewicz inequality with constant $\mu > 0$: for all $\mathbf{x} \in \mathbb{R}^n$,

$$\frac{1}{2}\|\nabla f(\mathbf{x})\|^2 \geq \mu(f(\mathbf{x}) - f^*),$$

where $f^* = \min_{\mathbf{x}} f(\mathbf{x})$ is the global minimum value.

Assumption 3 (Existence of Minimizer): The function f attains its minimum at some $\mathbf{x}^* \in \mathbb{R}^n$, and $f^* = f(\mathbf{x}^*) > -\infty$.

4.1.2 Lemmas

Lemma 1 (One-Step Progress Inequality): For any iterate $\mathbf{x}_k$ and coordinate i_k , the update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \frac{1}{L_{i_k}} \nabla_{i_k} f(\mathbf{x}_k) \mathbf{e}_{i_k}$$

satisfies:

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - \frac{1}{2L_{i_k}} (\nabla_{i_k} f(\mathbf{x}_k))^2.$$

Proof: Apply coordinate-wise smoothness (Assumption 1) with $t = -\frac{1}{L_{i_k}} \nabla_{i_k} f(\mathbf{x}_k)$:

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) + t \nabla_{i_k} f(\mathbf{x}_k) + \frac{L_{i_k}}{2} t^2.$$

Substitute t :

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - \frac{1}{L_{i_k}} (\nabla_{i_k} f(\mathbf{x}_k))^2 + \frac{L_{i_k}}{2} \left(\frac{1}{L_{i_k}^2} (\nabla_{i_k} f(\mathbf{x}_k))^2 \right).$$

Simplify:

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - \frac{1}{2L_{i_k}} (\nabla_{i_k} f(\mathbf{x}_k))^2.$$

□

Lemma 2 (Expected Descent Lemma): The expected function value at iteration $k+1$ satisfies:

$$\mathbb{E}[f(\mathbf{x}_{k+1}) \mid \mathbf{x}_k] \leq f(\mathbf{x}_k) - \frac{1}{2n} \sum_{i=1}^n \frac{1}{L_i} (\nabla_i f(\mathbf{x}_k))^2.$$

Proof: Take expectation of Lemma 1 over $i_k \sim \text{Uniform}(\{1, \dots, n\})$:

$$\mathbb{E}[f(\mathbf{x}_{k+1}) \mid \mathbf{x}_k] \leq f(\mathbf{x}_k) - \frac{1}{2n} \sum_{i=1}^n \frac{1}{L_i} (\nabla_i f(\mathbf{x}_k))^2.$$

□

Lemma 3 (Lower Bound on Gradient Sum): Let $L_{\max} = \max_i L_i$. Then:

$$\sum_{i=1}^n \frac{1}{L_i} (\nabla_i f(\mathbf{x}))^2 \geq \frac{1}{L_{\max}} \|\nabla f(\mathbf{x})\|^2.$$

Proof: Since $L_i \leq L_{\max}$, we have $\frac{1}{L_i} \geq \frac{1}{L_{\max}}$. Multiply both sides by $(\nabla_i f(\mathbf{x}))^2 \geq 0$ and sum over i :

$$\sum_{i=1}^n \frac{1}{L_i} (\nabla_i f(\mathbf{x}))^2 \geq \frac{1}{L_{\max}} \sum_{i=1}^n (\nabla_i f(\mathbf{x}))^2 = \frac{1}{L_{\max}} \|\nabla f(\mathbf{x})\|^2.$$

□

Lemma 4 (PL Inequality Application): Under Assumption 2,

$$\|\nabla f(\mathbf{x})\|^2 \geq 2\mu (f(\mathbf{x}) - f^*).$$

Proof: Multiply both sides of the PL inequality by 2:

$$\|\nabla f(\mathbf{x})\|^2 \geq 2\mu (f(\mathbf{x}) - f^*).$$

□

4.2 Main Proof (Step-by-Step Derivation)

We analyze the expected function value gap $\mathbb{E}[f(\mathbf{x}_k) - f^*]$.

4.2.1 Step 1: One-Step Progress

From Lemma 1:

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - \frac{1}{2L_{i_k}} (\nabla_{i_k} f(\mathbf{x}_k))^2.$$

4.2.2 Step 2: Take Expectation Over i_k

Using Lemma 2:

$$\mathbb{E}[f(\mathbf{x}_{k+1}) \mid \mathbf{x}_k] \leq f(\mathbf{x}_k) - \frac{1}{2n} \sum_{i=1}^n \frac{1}{L_i} (\nabla_i f(\mathbf{x}_k))^2.$$

4.2.3 Step 3: Apply Lemma 3

Substitute Lemma 3 into the inequality:

$$\mathbb{E}[f(\mathbf{x}_{k+1}) \mid \mathbf{x}_k] \leq f(\mathbf{x}_k) - \frac{1}{2nL_{\max}} \|\nabla f(\mathbf{x}_k)\|^2.$$

4.2.4 Step 4: Apply Lemma 4 (PL Condition)

Substitute $\|\nabla f(\mathbf{x}_k)\|^2 \geq 2\mu(f(\mathbf{x}_k) - f^*)$:

$$\mathbb{E}[f(\mathbf{x}_{k+1}) \mid \mathbf{x}_k] \leq f(\mathbf{x}_k) - \frac{\mu}{nL_{\max}} (f(\mathbf{x}_k) - f^*).$$

4.2.5 Step 5: Subtract f^* and Take Total Expectation

Subtract f^* from both sides:

$$\mathbb{E}[f(\mathbf{x}_{k+1}) - f^* \mid \mathbf{x}_k] \leq \left(1 - \frac{\mu}{nL_{\max}}\right) (f(\mathbf{x}_k) - f^*).$$

Take total expectation:

$$\mathbb{E}[f(\mathbf{x}_{k+1}) - f^*] \leq \left(1 - \frac{\mu}{nL_{\max}}\right) \mathbb{E}[f(\mathbf{x}_k) - f^*].$$

4.2.6 Step 6: Unroll the Recursion

Iterate the inequality:

$$\mathbb{E}[f(\mathbf{x}_k) - f^*] \leq \left(1 - \frac{\mu}{nL_{\max}}\right)^k (f(\mathbf{x}_0) - f^*).$$

□

4.3 Rate Derivation (With Constants)

The convergence rate factor is $\rho = 1 - \frac{\mu}{nL_{\max}}$. Key observations: 1. **Dependence on n :** The rate deteriorates linearly with the problem dimension n , reflecting the cost of updating only one coordinate per iteration. 2. **Dependence on $L_{\max}$:** Larger coordinate smoothness constants slow convergence, analogous to the condition number L/μ in gradient descent. 3. **Comparison to Full Gradient Descent:** Full gradient descent under PL achieves rate $1 - \frac{\mu}{L}$, where L is the global smoothness constant. Coordinate descent incurs an additional n -factor in the denominator.

4.4 Special Cases

4.4.1 Case 1: Uniform Coordinate Smoothness

If $L_i = L$ for all i , then $L_{\max} = L$, and the rate becomes:

$$\mathbb{E}[f(\mathbf{x}_k) - f^*] \leq \left(1 - \frac{\mu}{nL}\right)^k (f(\mathbf{x}_0) - f^*).$$

4.4.2 Case 2: Full Gradient Descent ($n = 1$)

For $n = 1$, coordinate descent reduces to gradient descent with step size $1/L$. The rate becomes:

$$\left(1 - \frac{\mu}{L}\right)^k,$$

matching the classical PL gradient descent rate.

4.4.3 Case 3: Strongly Convex Functions

If f is μ -strongly convex, it satisfies the PL condition with the same μ . The derived rate matches the known linear convergence rate for coordinate descent under strong convexity.

4.4.4 Case 4: Separable Functions

For $f(\mathbf{x}) = \sum_{i=1}^n f_i(x_i)$, coordinate descent decouples into independent scalar minimizations. The PL condition on f implies PL conditions on each f_i , preserving linear convergence.

This completes the rigorous derivation of linear convergence for randomized coordinate descent under the Polyak-Łojasiewicz condition and coordinate-wise smoothness.